Mengxuan Shen

MSc Robotics | University of Manchester

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Education

The University of Manchester
MSc Robotics *2026–2027*

Xi'an Jiaotong-Liverpool University
BEng Digital Media Technology, ***First-Class Honours*** *Aug. 2022–Jun. 2026*

- Relevant modules: Artificial Intelligence, Image Processing, Embedded Computer Systems, Microprocessor Systems, Computer Graphics, Signals and Systems, Electronic Circuits, Mobile Computing, Human-Centric Computing.
- Final-year thesis: *Multi-Head Learning for Computation Offloading in Mobile Edge Computing* (see Research).

Internship Experience

Vision System Assistant, Hebei Beifang Welding Tools Co., Ltd.
Jul. 2025–Sep. 2025

- Addressed a quality-inspection bottleneck where manual checking of robotic welding seams took 3–5 minutes per electrode holder; proposed an image-based pre-screening tool to flag obvious defects before human verification.
- Collected 450 weld images with the workshop's industrial camera and wrote Python/OpenCV scripts to extract seam regions using edge detection and morphological operations; labelled crack, porosity and incomplete-fusion patterns as ground truth.
- With a senior engineer, implemented a defect-detection pipeline combining Canny edge detection (cracks) and contour analysis (porosity), tuning thresholds with feedback from two QC inspectors across 120 test cases.

Defect Analysis Intern, Hebei Jincheng Yuan Electric Power Tools Co., Ltd.
Jul. 2024–Oct. 2024

- Investigated a 30% false-alarm rate in a newly deployed automated inspection system for insulated tool handles by analysing 200 misclassified images; used histogram analysis to quantify brightness differences between true defects and false positives.
- Built a Python/OpenCV preprocessing pipeline comparing Gaussian, median and bilateral filtering; adaptive histogram equalisation with morphological closing reduced lighting-induced false alarms by 45% in controlled tests on 150 samples.

Research Experience

Multi-Head Learning for Edge Computing Offloading
Final-Year Thesis, Xi'an Jiaotong-Liverpool University *2025–2026*

- Extended a shared-backbone multi-task network (MTFNN-CO baseline) with five specialised prediction heads to decide which mobile tasks to offload to an edge server (Python, TensorFlow/Keras).
- Designed an "enhanced gating" head selector using only deployment-available signals; it outperformed voting and input-only gating at every scale tested (e.g. 82.4% vs 79.1% decision accuracy at four users).

Vision-Guided Robotic Control for Hand Rehabilitation
Research Assistant, Robotics Laboratory *Oct. 2024–Feb. 2025*

- Developed a real-time visual feedback system for a rehabilitation robot that previously followed fixed trajectories, detecting hand keypoints with OpenCV and MediaPipe from an overhead camera.
- Designed a position-based visual servoing controller in Python, mapping joint coordinates from image space to the robot frame through calibration matrices to adjust trajectories to the patient's actual position.
- Integrated visual feedback into the control loop so the robot adapted when subjects shifted posture, reducing the need for manual therapist intervention.

Deep Learning for Medical Image Enhancement
Research Assistant, Medical Imaging Group *Feb. 2024–Jun. 2024*

- Built a U-Net enhancement model for degraded chest X-rays (poor contrast, motion blur, under-exposure), with a weighted loss emphasising anatomical boundaries and suspected abnormal regions, plus elastic and intensity augmentation.

Edge Machine Learning for IoT Navigation System
Research Assistant, Embedded Systems Lab *Jan. 2024–May 2024*

- Developed an on-device decision-tree model for a smart umbrella to recommend routes from weather patterns without cloud dependency, using GPS and environmental sensor data, deployable on Arduino hardware.

Retrieval-Augmented Generation for Research Question Answering
Research Assistant under Prof. Philipp Koehn (Johns Hopkins University) *May 2023–Sep. 2023*

- Built a hybrid retrieval pipeline (BM25 filtering, then Sentence-BERT embeddings with a FAISS index for semantic re-ranking) to cut query response time while keeping relevant papers in the top results.

Project Experience

Intelligent Umbrella for Rainy-Day Navigation
Team Leader, Jiangsu Provincial College Robot Creativity Competition — ***Second Prize*** *Mar. 2025–Jun. 2025*

- Led a team of four to prototype a smart umbrella with LED lighting, NFC, GPS tracking and 4G connectivity; wrote Arduino C/C++ for NFC data exchange, GPS acquisition and 4G communication.
- Designed structural parts in CAD and 3D-printed iterative prototypes; managed task allocation, troubleshooting and the competition demonstration.

AI Personal Assistant Mobile Application (AIVA)

UI/UX Designer *Mar. 2025–May 2025*

- Designed the full interface for an AI assistant app (scheduling, diet recommendations, decision support) in Figma, from wireframes to high-fidelity interactive prototypes.

Other Projects

Source code and demos: shine574269.github.io *2024–2026*

- DailySense Android app (UI design and calendar module); TED Talks TF-IDF search and recommendation system; Unity 2D shooter (C#); 8051 embedded pulse counter (C); OpenGL graphics (C++); MATLAB image processing and eigenface recognition.
- Mathematical Contest in Modeling: led a three-member team through a 96-hour optimisation problem using Python, Java and MATLAB.

Awards

- Second Prize, Jiangsu Provincial College Intelligent Robot Creativity Competition (Jul. 2025)
- Honorable Mention, Mathematical Contest in Modeling (MCM) (Feb. 2024)
- First Place, ELC Sustainable Development Video Creation Competition (Apr. 2023)
- Provincial Outstanding Class, Xi'an Jiaotong-Liverpool University (Jun. 2023)

Skills

- **Programming:** Python, Java (proficient); C/C++, Embedded C, C#
- **AI & Computer Vision:** OpenCV, MediaPipe, PyTorch, TensorFlow; CNNs, U-Net, Transformers; visual servoing
- **Embedded & Hardware:** Arduino, 8051 (Keil), GPS/4G/NFC modules, CAD and 3D printing, MATLAB
- **Tools & Design:** Git, MySQL, Unity, Android Studio, Figma, Photoshop, Premiere

Leadership & Activities

- Director of Administration & Finance, Student Union, XJTLU (Feb. 2023–Jun. 2024); Head of Publicity, International Innovation Society (Feb.–Nov. 2024); Class Monitor (2022–2023)
- Translator Volunteer, International Wushu Invitational Tournament (2024); All-Subject Tutor (2022–2025)